

# TASK 4 – REMOTE STATE MANAGEMENT & TERRAFORM MODULES

## *Terraform-Managed Azure Subscription Baseline*

### 1. Objective

To improve the maintainability and collaboration model of the Terraform project by moving Terraform state to an Azure Storage backend and refactoring the existing infrastructure into reusable modules. This task covers Week 7 and Week 8.

### 2. Week 7 – Remote Terraform State

Terraform state was migrated from local state to an Azure Storage backend. The backend uses the storage account **stterraformbaseline01**, the **tfstate** container and the key **terraform.tfstate**. Azure AD authentication is used for backend access. The signed-in user was granted Storage Blob Data Contributor permission for the state storage account.

After migration, Terraform successfully acquired and released the remote state lock and reported that the deployed infrastructure matched the configuration. This provides centralized state storage and supports safer concurrent Terraform operations.

### 3. Week 8 – Reusable Terraform Modules

The existing working infrastructure was refactored into four reusable modules: **network**, **compute**, **security** and **monitoring**. Each module separates its variables, resources and outputs, while the root configuration supplies module inputs and consumes module outputs.

Terraform moved blocks were used to migrate existing resource addresses into the new module addresses. This allowed the project structure to change without destroying and recreating the already deployed Azure resources.

### 4. Architecture

| Module     | Main Responsibility                                       |
|------------|-----------------------------------------------------------|
| network    | VNet, subnet, NSG, HTTP/SSH rules and NSG association     |
| compute    | Public IP, NIC and Linux virtual machine                  |
| security   | User-assigned managed identity and Reader RBAC assignment |
| monitoring | Log Analytics workspace and VM diagnostic setting         |

### 5. Resource Migration with moved Blocks

The moved.tf configuration maps the original root-level resource addresses to their new module addresses. During migration, Terraform reported zero resources to add, change or destroy. This verifies that the refactoring changed the Terraform state structure without recreating the Azure resources.

### 6. Expected Result

The project uses a remote Azure backend for Terraform state and a modular code structure for networking, compute, security and monitoring. The infrastructure remains deployed while the Terraform configuration becomes easier to reuse, maintain and extend.

### 7. Evidence

The accompanying README contains the recommended screenshot checklist. Screenshots should demonstrate the Azure Storage backend, remote state, state locking, module structure, moved blocks and successful zero-change migration. Do not include access credentials or private keys.